USN | | | | | | | | | | | [24MTAI13]

FIRST SEMESTER M.Tech EXAMINATION, JANUARY – 2025

MACHINE LEARNING

Time: 3 Hours **Maximum Marks: 100**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 20 marks.
ii. Missing data may be suitably assumed.

			CO's	Blooms Level	Marks
1.	**a.**	Describe machine learning techniques with examples and illustrate their applications and differences.	**1**	**2**	**10**
	b.	Explain the concept of Mean Squared Logarithmic Error (MSLE) with an example to illustrate how MSLE is calculated and interpreted.	**1**	**2**	**10**
2.	**a.**	Analyse the working mechanism of Random Forests as an ensemble learning method.	**2**	**4**	**10**
	b.	Analyse the Bootstrap method in sampling and how it differs from traditional sampling techniques.	**2**	**4**	**10**
3.	**a.**	Illustrate any two Value-Based Reinforcement Learning methods with an example.	**3**	**2**	**10**
	b.	Discuss the differences between Supervised, Unsupervised and Reinforcement Learning.	**3**	**2**	**10**
		OR			
4.	**a.**	Compare Q-learning and distributional DQN.	**3**	**4**	**10**
	b.	Examine various components to learn a policy of Reinforcement learning.	**3**	**4**	**10**
5.	**a.**	Explain the concept of hyperparameter fine-tuning in machine learning models to improve model performance.	**4**	**3**	**10**
	b.	Apply resampling techniques in statistics to estimate the sampling distribution of a statistic.	**4**	**3**	**10**
		OR			
6.	**a.**	Examine the evaluation metrics based on Precision, Recall, Accuracy, Specificity, and F1 Score in the context of model performance assessment.	**4**	**4**	**10**
	b.	Examine any two ranking metrics used to evaluate the performance of models that produce ranked outputs.	**4**	**4**	**10**

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7.	**a.**	Outline and explain the Python code snippet that demonstrates how to connect to a SQLite database using the sqlite3 module.	**5**	**3**	**10**
	b.	Analyse the process of model serialization in machine learning and explain how the 'pickle' and 'joblib' libraries are used for deploying serialization in Python.	**6**	**4**	**10**
		OR			
8.	**a.**	How would you define the scope of a regression problem when developing a web application? Discuss the key factors that should be considered to ensure the application effectively meets user needs.	**5**	**4**	**10**
	b.	Demonstrate the steps involved in developing a web application using Flask.	**6**	**4**	**10**